

Ultra-Detailed Schema & Theoretical Guidance: Next-Gen Diabetes Risk

This document serves as the research and control specification (Phase 1, Steps 3 & 4) for generating the 500-row "Golden Seed" dataset.

Part 1: Theoretical Foundation (The "Why")

This section explains the clinical relevance of each feature and how they are expected to interact based on established medical knowledge. This is the context the generator LLM needs to ensure clinical plausibility.

Feature	Clinical Significance & Relation to T2D	Importance Rank
Age	Type 2 Diabetes (T2D) risk is highly cumulative, increasing sharply after age 40 due to decreased insulin sensitivity and metabolic changes. The model must show a strong, linear correlation: Older Age $\implies$ Higher Risk/Diagnosis.	High
BMI	Body Mass Index is a primary driver of insulin resistance. It must correlate directly with Fasting_Glucose_Level . High BMI $\implies$ High Glucose/Diagnosis. The model should also reflect that a high BMI can be mitigated by high Exercise_Frequency .	High
Fasting_Glucose_Level	This is the diagnostic measure. Its value <i>directly defines</i> the Diabetes_Diagnosis target variable based on established thresholds. It is also the feature most influenced by the <i>combination</i> of all other risk factors (Age, BMI, Genetics).	Critical
Genetic_Risk_Score & Family_History	These represent non-modifiable, foundational risk. They should act as a risk multiplier . A high genetic score/history alone should lead to 'Pre-Diabetic' status, but only when combined with poor lifestyle (High BMI, Sedentary) should it push an individual into 'Type 2 Diabetic' status.	Medium
Exercise_Frequency	Acts as a risk mitigator . It must show an inverse correlation with glucose and diagnosis risk. Individuals with 'High' exercise should have lower BMI and lower glucose levels than genetically similar, sedentary individuals.	Medium
Target Variable: Diabetes_Diagnosis	Defined by the Fasting_Glucose_Level thresholds: Negative (< 100 mg/dL), Pre-Diabetic (100–125 mg/dL), Type 2 Diabetic ($\geq$ 126 mg/dL).	Critical

Part 2: Statistical Targets (The "What Numbers")

This section defines the numerical goals and relationships that must be maintained in the final 500-row dataset. These metrics will be used for the iterative validation loop (Phase 2, Step 7).

A. Univariate (Single Feature) Targets

Feature	Data Type	Range/Categories	Target Mean/Distribution	Guidance & Why
Age	Integer	$25 \leq x \leq 75$	Mean: ≈ 50	We need the mean skewed older to ensure a high-risk population for model training.
BMI	Float	$18.5 \leq x \leq 45.0$	Mean: ≈ 29.0	We need the mean skewed into the "Overweight/Obese" category (≥ 25) to generate enough 'Pre-Diabetic' and 'Diabetic' cases.
Fasting_Glucose_Level	Float	$70.0 \leq x \leq 300.0$	Mean: ≈ 115.0	The mean must fall into the 'Pre-Diabetic' range (100-125) to ensure sufficient representation of all three classes.
Genetic_Risk_Score	Integer	$1 \leq x \leq 10$	Mean: ≈ 6	Skewed slightly higher to ensure the genetic risk factor is present across the dataset.
Exercise_Frequency	Categorical	'Sedentary', 'Low', 'Moderate', 'High'	Distribution: {Low: 30%, Sedentary: 30%, Moderate: 25%, High: 15%}	We need a higher proportion of Sedentary/Low to keep the risk pool high, but enough Moderate/High to show mitigation.
Diabetes_Diagnosis (Target)	Categorical	'Negative', 'Pre-Diabetic', 'Type 2 Diabetic'	Target Distribution: {Negative: 35%, Pre-Diabetic: 35%, Type 2 Diabetic: 30%}	This distribution ensures a balanced dataset (avoiding the real-world imbalance) for effective model training. This is a non-negotiable target.

B. Multivariate (Feature Relationship) Targets

These correlations are crucial for the model to learn the correct predictive relationships.

Relationship Pair	Target Correlation (Pearson's ρ)	Guidance & Why
Age vs. Fasting_Glucose_Level	$\rho \geq 0.55$ (Strong Positive)	The primary relationship: Risk must rise significantly with age.
BMI vs. Fasting_Glucose_Level	$\rho \geq 0.60$ (Strong Positive)	The secondary relationship: Weight gain must be the strongest driver of T2D.

Genetic_Risk_Score vs. Fasting_Glucose_Level	$\rho \approx 0.30$ (Weak to Moderate Positive)	Genetic risk is important but not the <i>strongest</i> factor; lifestyle factors (BMI/Exercise) must dominate.
Exercise_Frequency vs. BMI	$\rho \approx -0.40$ (Moderate Negative)	Mitigation Check: The data must show that high exercise correlates with lower BMI.

C. Logical Constraints (Hard Rules)

The LLM must adhere to these absolute rules, which the RAG validator (and the SME) will check for:

- 1. **Diagnosis Rule:** Any row where `Fasting_Glucose_Level` is ≥ 126 **MUST** have `Diabetes_Diagnosis` = 'Type 2 Diabetic'.
- 2. **Health Paradox Rule:** Individuals with `Exercise_Frequency` = 'High' and `BMI` < 25 should almost always have `Fasting_Glucose_Level` < 100 , **regardless** of `Age` or `Genetic_Risk_Score` . (Extremely high genetic score may be the rare exception).
- 3. **Risk Floor Rule:** Individuals with `Age` < 40 , `BMI` < 25 , and `Genetic_Risk_Score` ≤ 3 **MUST** have `Diabetes_Diagnosis` = 'Negative'.

Next Step

The Theoretical Guidance and Statistical Targets are now complete and structured. The next step is to begin the iterative generation process (Phase 2, Step 5) and create the first single-row sample.